BOXSPLOIT LABS / PRODUCT DOCUMENTATION

# PANBOX

## Product & operator guide

Self-hosted AppSec control plane. Setup, CI connectivity, access control, operations, and honest pilot limits.

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This guide describes implemented pilot behavior. Roadmap items are labelled separately.

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Start with the path that matches your job: server operator installs the private dashboard once; DevOps adds an agent-only step to each CI job. The agent is one Go binary. The runner does not need Node, Python, Semgrep, Grype, or a PANBOX server installation.

# 1. What runs where

```
Git host/CI -> protected runner -> PANBOX agent -> private PANBOX API -> PostgreSQL
                                  | VACCIN: source rules            |
                                  | ENETRON: packages -> OSV        +-> React UI
```

Result: CI receives pass, gate failure, incomplete, or authorization failure; operators see repository and branch snapshots. Source files stay on the runner. ENETRON sends package names and versions to an OSV-compatible advisory API. The agent uploads normalized findings and a repository-bound scan key to PANBOX, without raw source snippets. See the architecture and network diagram.

# 2. Choose your setup path

- Server operator, once: Docker Engine/Desktop with Linux containers and Compose v2. Start with section 3 on Debian/Linux/macOS or section 4 on Windows. Debian is the initial production target.
- DevOps, per repository: enroll a repository, create one scan key, set CI secrets, and run section 5. The runner needs outbound HTTPS to GitHub Releases and the advisory API, plus a route to the private PANBOX API.
- No public IP/domain: choose a route in section 6 before writing the CI job.

PANBOX binds HTTP 8080 and PostgreSQL 5434 to host loopback by default. Change PANBOX_HOST_PORT or PANBOX_DB_HOST_PORT in .env if needed. Keep PostgreSQL private.

# 3. Start the dashboard on Debian/Linux/macOS

Goal: bring up the private API and UI once. Run on the server host:

```
git clone https://github.com/alterwriter/PANBOX.git
cd PANBOX
sh scripts/init-local.sh
docker compose --env-file .env -f deploy/docker-compose.yml up -d --build
curl -fsS http://127.0.0.1:8080/readyz
```

Expected: /readyz succeeds. Open http://127.0.0.1:8080 on the server and sign in with PANBOX_BOOTSTRAP_EMAIL and PANBOX_BOOTSTRAP_PASSWORD from the ignored .env. The initializer creates secrets with restricted permissions and refuses to overwrite an existing .env. Bootstrap credentials apply only when the database is first initialized; retain its volume on routine restarts.

Optional demo: PANBOX_SERVER=http://127.0.0.1:8080 bash scripts/seed-demo.sh needs curl and jq, then prompts for the admin password. Demo findings are synthetic.

# 4. Start the dashboard on Windows

Goal: run the same Linux containers through Docker Desktop. In PowerShell:

```
git clone https://github.com/alterwriter/PANBOX.git
cd PANBOX
./scripts/init-local.ps1
docker compose --env-file .env -f deploy/docker-compose.yml up -d --build
Invoke-RestMethod http://127.0.0.1:8080/readyz
```

Expected: the same dashboard and API as section 3. Use ./scripts/seed-demo.ps1 only for optional demo data. Run one initializer for a new .env, not both.

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# 5. Add only the agent to CI

Once in PANBOX UI: Repositories -> enroll a GitHub, GitLab, Bitbucket, Azure Repos, Gitea, or Other Git repository using its stable provider ID/path. Copy the internal PANBOX repository ID. Generate its scan key. Store it as protected CI secret PANBOX_TOKEN; set PANBOX_REPOSITORY_ID and a reachable private PANBOX_SERVER CI variable. Never put a scan key in repository files or untrusted pull request jobs.

Every Linux/macOS job: download the pinned installer from PANBOX release v0.1.0-alpha.4. It selects OS/CPU, downloads just the binary, checks SHA256SUMS, and installs in the job workspace without sudo:

```
PANBOX_VERSION=v0.1.0-alpha.4
curl -fsSL
"https://github.com/alterwriter/PANBOX/releases/download/${PANBOX_VERSION}/install-agent.sh" -o
install-panbox-agent.sh
sh install-panbox-agent.sh "$PANBOX_VERSION" .panbox/bin
.panbox/bin/panbox go --server "$PANBOX_SERVER" --repo "$PANBOX_REPOSITORY_ID" --path .
```

Expected: exit 0 complete/pass; 1 gate failure; 2 incomplete/transport/config error; 3 rejected auth/permission. Copy the GitHub Actions job or GitLab CI job. On Windows, download install-agent.ps1 and run ./install-agent.ps1 -Version v0.1.0-alpha.4; it installs ./.panbox/bin/panbox.exe after checksum verification. Direct agent-only binaries are also available.

VACCIN checks selected Python, JavaScript/TypeScript, and Go source patterns. ENETRON matches pinned npm lockfiles v2/v3, requirements.txt, and go.mod dependencies against OSV advisory data. OSV supplies data, not scanner execution. PANBOX_ADVISORY_URL or --advisory-url selects an approved OSV-compatible mirror. --offline suppresses dashboard upload, but ENETRON still contacts the advisory API unless only --modules vaccin is selected.

# 6. Reach private PANBOX from public SCM

- Local protected runner (recommended): polls any SCM over outbound HTTPS and posts to PANBOX on its LAN. No public PANBOX port; synchronous gate. Use the GitHub or GitLab example.
- SSH relay: PANBOX opens an outbound reverse tunnel to a reachable relay; a shared runner opens a forward tunnel. No PANBOX inbound internet port; synchronous gate. Pin host keys and restrict identities. See the GitLab relay example.
- Artifact import: CI writes panbox-results.json; a private host fetches and imports it later. No inbound route or relay; the later gate cannot block that CI job. GitLab artifact example and panbox relay-pull automate GitLab; other SCMs need their artifact download step before generic panbox go --bundle import.

Check: from a synchronous runner, curl -fsS "$PANBOX_SERVER/readyz" must work. A public repository does not make its CI artifacts public.

# 7. Give people only needed access

Admin setup: Settings & Audit -> RBAC -> create a role with explicit permissions -> assign users -> grant repository membership. admin is the highest immutable role. Built-in roles include security_reviewer, developer, executive_viewer, and scanner_ci; custom roles cannot receive rbac:manage.

Check: data access needs both permission and repository membership. Pilot permissions are repos:read, repos:write, scans:read, scans:write, findings:read, reports:export, rbac:manage, tokens:manage, and audit:read.

# 8. Create API, access, and scan keys

Where: Settings & Audit -> API & access keys. Choose label, type, expiry, scopes, and repository when applicable. API keys carry selected integration scopes; access keys are read-only; scan keys have scans:write for one repository. Copy the one-time secret into a secret manager. Keys are hashed at rest, expiring, and revocable.

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Read-only smoke test:

```
curl -H "Authorization: Bearer $PANBOX_ACCESS_KEY" "$PANBOX_SERVER/v1/repos"
```

Browser login uses password, HttpOnly session cookie, and CSRF token. The OpenAPI contract lists pilot endpoints. Rotate by creating a new key, updating the client, confirming access, then revoking the old key.

## 9. Confirm the scan in the dashboard

After CI: Overview shows coverage and severity; Repositories shows enrolled projects; Scans shows run/gate status; Findings filters by repository/branch/module/severity; Reports exports technical CSV/JSON; Settings & Audit shows RBAC, keys, and audit events. Light and dark themes are available.

Interpretation: a partial scan is unverified and preserves the last successful posture. ARK labels Critical/High and BUG-STER labels Medium/Low while standard severities remain stored and exported. Interactive recasting is not implemented in this pilot.

## 10. Operate within the pilot boundary

Keep PostgreSQL on loopback. For remote browser access, use an HTTPS reverse proxy with access control or a private route; set PANBOX_SECURE_COOKIE=true for HTTPS. Back up PostgreSQL and test restoration.

The 2 GB Debian host is a target to validate, not a guarantee. A small Windows Docker Desktop fixture measured about 5.1 MiB server RAM and 37.3 MiB PostgreSQL RAM; 100 sequential API calls measured 7.1 ms median and 9.3 ms p95. The agent runs on the runner and needs separate sizing. Measure docker stats --no-stream with representative repositories and scan bundles.

## 11. Troubleshoot by failed step

- Dashboard not ready: run docker compose --env-file .env -f deploy/docker-compose.yml ps and logs db server; confirm PostgreSQL health.
- Agent download fails: check release tag, OS/CPU asset name, outbound GitHub access, curl, and SHA-256 tooling. Do not skip the checksum.
- Runner cannot reach PANBOX: 127.0.0.1 inside a container is that container; use its reachable private host/network address or the SSH relay.
- Exit 2: check manifest, advisory endpoint, work path, PANBOX endpoint, and logs; incomplete is not pass.
- Exit 3: check scan key expiry/revocation, scans:write, internal repository ID, role, and membership.
- Artifact import is late: check timer, GITLAB_READ_TOKEN, allowed refs, Jobs API/artifact access, and relay state file.

## 12. Know the alpha limits

Available now: server/UI, native VACCIN source checks, native ENETRON pinned dependency matching with OSV data, repository/branch snapshots, gate, RBAC, scoped keys, audit, CSV/JSON exports, and the transport examples above. Validate real runners, OSV access or approved mirror, Debian load, backup/restore, TLS, and key rotation in your environment.

Coverage is intentionally narrow. RIDER secrets, TESLA IaC, severity recasting/approval, finding/executive PDF reports, AI Security, MCP, OIDC, and API rate limiting are roadmap items. This PDF is a product guide, not a PANBOX vulnerability report.

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